Project Report: Responsive Contact Form

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1. Project Overview

The *Responsive Contact Form* is a frontend development project created as part of *the Elevvo Pathways Frontend Development Internship*. The project focuses on building a clean and responsive contact form that allows users to enter their personal information and send a message through a simple and organized interface.

The form includes fields for Full Name, Email Address, Phone Number, Subject, Message, Preferred Contact Method, and a consent checkbox. It also provides Send Message and Reset Form buttons.

The project was developed using HTML5, CSS3, and vanilla JavaScript. The main focus was on creating a responsive layout, styling the form clearly, and adding client-side validation to provide useful feedback when users enter incorrect or incomplete information.

2. Project Objective

The main objective of this project was to develop a clean and responsive contact form that works properly across different screen sizes while providing basic input validation.

The main goals were to:

- Build the form structure using semantic HTML5 elements.
- Create a clean layout with properly spaced form fields.
- Make the form responsive for both mobile and desktop screens.
- Use JavaScript to validate user input before submission.
- Display clear error messages when invalid information is entered.
- Validate required fields such as name, email, subject, and message.
- Validate the email format and optional phone number.
- Require user consent before allowing successful submission.
- Provide a success message when all validation requirements are satisfied.

These objectives follow the requirements of the task, which focuses on HTML forms, CSS styling, responsive design, layout, and basic JavaScript validation.

3. Development Process

I started the project by creating the basic contact form structure using HTML5. The form was divided into logical sections using <fieldset> and <legend> elements. The first section contains contact information, including the user's name, email address, and optional phone number.

I then created a separate message section containing a subject dropdown and a message textarea. I also added a preferred contact method section where users can choose between email and phone. A consent checkbox was included so users can confirm that they agree to the collection and use of their information.

After completing the HTML structure, I used CSS3 to style the form. I created a centered white form container with rounded corners, shadows, spacing, and a soft background gradient. The input fields were given consistent sizing, spacing, borders, and focus effects to make the form easier to use.

For responsive behavior, I added a media query for screens 768px and wider. On smaller screens, the form uses a simple single-column layout, while on larger screens the container receives additional spacing and the two buttons are displayed side by side.

Finally, I added JavaScript to handle form validation and user interaction. The script checks the user's input, displays appropriate error messages, and shows a success message when all required information is valid.

4. Core Features

The contact form includes several features designed to make it simple and easy to use.

- **Responsive Design:** The form adapts to different screen sizes using CSS media queries.
- **Contact Information:** Users can enter their full name, email address, and optional phone number.

- **Subject Selection:** Users can select a subject such as Job Opportunity, Project Collaboration, Feedback, or Other.
- **Message Field:** Provides a textarea for users to write their message.
- **Preferred Contact Method:** Allows users to choose between Email and Phone.
- **Consent Checkbox:** Requires users to agree before successfully submitting the form.
- **Email Validation:** Checks whether the entered email follows a valid format.
- **Phone Validation:** Allows optional phone input and checks for digits with a maximum of 11 digits.
- **Message Validation:** Requires the message and checks that it contains at least 10 characters.
- **Error Messages:** Displays specific validation messages below the relevant fields.
- **Success Message:** Displays a confirmation message after all validation requirements are met.
- **Reset Button:** Clears the form and removes validation messages.
- **Focus Effects:** Provides visual feedback when users interact with form controls.
- **Accessible Structure:** Uses labels, fieldsets, legends, ARIA attributes, and descriptive text to improve usability.

5. Technical Problem-Solving

During development, I faced several challenges related to form validation, responsive layout, and providing clear feedback to users. I solved these problems by combining HTML structure, CSS styling, and JavaScript event handling.

- **Validating required fields:** I used JavaScript conditions to check whether the name, email, subject, and message fields were empty.

- **Validating email addresses:** I used a regular expression to check whether the entered email follows a valid format.
- **Validating phone numbers:** I created a separate validation pattern that allows only digits and limits the phone number to 11 digits.
- **Validating message length:** I added a minimum requirement of 10 characters so that very short messages are not accepted.
- **Handling consent:** The form checks whether the consent checkbox is selected before showing the success message.
- **Displaying validation errors:** I created a reusable showError() function that adds an invalid style and displays the appropriate error message.
- **Clearing errors:** A clearErrors() function removes previous error messages and invalid styles before a new validation attempt.
- **Preventing invalid submission:** The submit event uses preventDefault() so the form can be checked with JavaScript before displaying the success message.
- **Responsive layout:** I used a CSS media query to change the spacing and button arrangement on larger screens.
- **Improving usability:** Focus effects, clear labels, error messages, and descriptive instructions were added to make the form easier to understand and interact with.

These solutions helped me understand how frontend technologies can work together to create a form that is not only visually organized but also responsive and interactive.

6. Technical Stack Summary

Technology	*Purpose*
HTML5	Used to create the structure of the contact form and organize the fields using semantic elements.
CSS3	Used for styling, spacing, colors, borders, focus effects, and responsive layout.
JavaScript	Used for form validation, DOM manipulation,

(ES6+)	event handling, and displaying feedback messages.
CSS Media Queries	Used to adapt the form layout for different screen sizes.
Regular Expressions	Used to validate email and phone number formats.
HTML Fieldsets & Legends	Used to logically group related form controls.
ARIA Attributes	Used to provide additional information for assistive technologies.
Visual Studio Code	Used to write and organize the project files.
Browser Developer Tools	Used to inspect and test the form during development.

7. Learning Outcomes

Working on this project gave me practical experience in creating and validating HTML forms using frontend technologies.

I improved my understanding of semantic HTML by working with form elements, labels, fieldsets, legends, radio buttons, checkboxes, select menus, and textareas.

The project also helped me understand how JavaScript can be used to interact with form elements through the DOM. I learned how to capture user input, validate different types of data, display error messages, and show a success message when the input is valid.

Another important learning outcome was responsive design. I learned how to create a form that maintains a clean layout on smaller screens while providing a more spacious arrangement on larger screens.

I also gained more experience with accessibility by using connected labels, fieldsets, legends, focus indicators, ARIA attributes, and descriptive instructions.

Overall, this project helped me strengthen my frontend development and problem-solving skills through a practical form-building task.

8. Conclusion

The **Responsive Contact Form** was a useful hands-on project that allowed me to apply HTML5, CSS3, and JavaScript to a practical frontend development task.

I created a structured contact form with responsive styling, multiple input types, client-side validation, clear error messages, a consent requirement, and a success message. The form was also designed to remain usable across different screen sizes.

Throughout the development process, I gained experience in form structure, CSS styling, responsive design, DOM manipulation, JavaScript validation, and accessibility considerations.

Overall, this project strengthened my understanding of how HTML, CSS, and JavaScript work together to create a functional and user-friendly web interface. It also gave me more confidence in building responsive frontend components and solving common form-related development problems.